

The outer positions were filled with “X,” which was never a memory item. After the stimulus, the letters disappeared from the screen, and the maintenance interval started ($[-3, 0]$ s). A fixation square was shown throughout fixation, encoding, and maintenance. After maintenance, a probe was presented. The subjects responded with a button press to indicate whether the probe was part of the stimulus. The subjects were instructed to respond as rapidly as possible without making errors. The hand used for the response was counterbalanced across subjects within the clinical constraints. After the response, the probe was turned off, and the subjects received acoustic feedback regarding whether their response was correct or incorrect. Before initiating the next trial, the subjects were encouraged to blink and relax. The subjects performed 50 trials in one session, which lasted approximately 10 min. Trials with different set sizes were presented in a random order, with the single exception that a trial with an incorrect response was always followed by a trial with a set size of 4. The task is freely available at www.neurobs.com/ex_files/expt_view?id=266. During the recording period of several days, several subjects decided to perform more than one session of the task up to a total of seven sessions.

SUBJECTS

Nine subjects participated in the study (table S1). All subjects were implanted with depth electrodes in the MTL for the potential surgical treatment of epilepsy. All subjects provided written informed consent for the study, which was approved by the institutional ethics review board (PB 2016-02055). All subjects had normal or corrected-to-normal vision and were right handed as confirmed by neuropsychological testing. The depth electrodes (1.3 mm diameter, 8 contacts of 1.6 mm length, and spacing between contact centers 5 mm; Ad-Tech, Racine, WI, www.adtechmedical.com) were stereotactically implanted into the amygdala, hippocampus, and entorhinal cortex. Each macroelectrode had nine microelectrodes that protruded approximately 4 mm from its tip. In addition, scalp EEG electrodes were placed at the sites of the 10-20 system with minor adaptations to avoid surgical scalp lesions.

Depth electrode localization

Electrodes were localized using postimplantation computed tomography (CT) scans and postimplantation structural T1-weighted MRI scans. For each subject, the CT scan was registered to the postimplantation scan as implemented in FieldTrip (35, 36). In the coregistered CT-MR images, the electrode contacts were visually marked. The contact positions were normalized to the MNI space and assigned to a brain region using the Brainnetome Atlas (37). In addition, depth electrode positions were verified by the neurosurgeon (L.S.) after merging preoperative MRI with postimplantation CT images of each individual subject in the plane along the electrode (iPlan Stereotaxy 3.0, Brainlab, München, Germany). We grouped electrodes according to whether they were recorded from the SOZ or another area. As an illustration, we projected the three-dimensional positions of all electrode tips, i.e., the positions of the microelectrodes, on a parasagittal plane (MNI space $x = -25.2$ mm; Fig. 1D).

Recording setup

Intracranial data were acquired at sampling frequencies of 4 kHz for the contacts of the macroelectrodes and 32 kHz for the microelectrodes via the ATLAS recording system (0.5- to 5000-Hz passband,

Neuralynx, Bozeman, MT, USA; www.neuralynx.com). iEEG was recorded (Neuralynx ATLAS; sampling rate, 4000 Hz; passband, 0.5 to 1000 Hz) against a common intracranial reference and subsequently transformed to a bipolar montage. Only one bipolar signal from a pair involving the deepest contacts of each macroelectrode was used for further analysis. Scalp EEG was recorded (NicOne; sampling rate, 256 Hz; passband, 0.3 to 100 Hz). Scalp EEG was re-referenced to the averaged mastoid channels.

Spike detection and single-unit identification

The Combinato package 17 (<https://github.com/jniediek/combinato>) was used for spike sorting (38). Combinato follows the same procedure of other freely available software packages: peak detection in the high-pass (>500 Hz) signal, computation of wavelet coefficients for detected peaks, and superparamagnetic clustering in the feature space of wavelet coefficients. As an advantage over other clustering procedures, Combinato provides better automated artifact rejection and is more sensitive in the detection of clusters of small size (few action potentials). We visually inspected each identified cluster based on the shape and amplitude of the action potentials and the interspike interval (ISI) distributions. We removed clusters that exhibited a low firing rate (<0.1 Hz), noisy waveforms, or nonuniform amplitude or shape of the action potentials in the recorded time interval. Moreover, to avoid overclustering, we merged highly similar clusters identified on the same microelectrode. Last, we computed several metrics of spike sorting quality (fig. S2): The percentage of ISI below 3 ms was $1.58 \pm 4.18\%$, and the ratio of the peak amplitude of the mean waveform to the SD of the noise was 3.28 ± 1.56 .

Statistics

For statistical significance analysis, we used nonparametric permutation tests. To assess the significance of a value, we created a null distribution estimated from $n > 200$ permutations on data with scrambled labels. The P value is limited by the number of permutations as $P = 1/(\text{number of permutations} + 1)$. Reported P values were based on the empirically estimated null distribution. If no value of the null distribution exceeded the observed value, we reported the exact P value.

To compare two distributions, we used a permutation t test wherein we randomly mixed values from the two distributions 2000 times to create a distribution of t values and computed an empirical P value from this distribution using the EEGLAB toolbox (39). For comparisons between more than two groups, we used F statistics. When comparing firing rates or power spectra, we used cluster-based nonparametric permutation tests (36). For each test, matrix sizes, number of samples, and features are provided in table S2.

dPCA of firing rates

To illustrate the firing rates of all neurons over the periods of a trial, we used dPCA as a dimensionality reduction method (24). In contrast to PCA, which explains the variance of components regardless of task conditions, dPCA incorporates the task conditions, i.e., set size in our task, when explaining the variance. We used nonoverlapping bins of 2 ms for neuronal firing and smoothed the rates in a window of 1000 ms with a Gaussian kernel. We z -scored the resulting rates based on the mean of the firing rate during fixation. We computed time-dependent and set size-dependent dPCA components and ordered them by explained variance. To test the significance of the explained variance, we created a null distribution of explained variance by scrambling labels

and computing dPCA 200 times. For each component, we compared the explained variance to the null distribution.

For the component trajectories in the dPC space, we defined multidimensional distance

$$d(p(t), q(t)) = \sqrt{\sum_{i=1}^n (p_i(t) - q_i(t))^2} \quad (1)$$

and multidimensional speed

$$V(t) = \frac{1}{n} \times \frac{\sqrt{\sum_{i=1}^n (p_i(t) - p_i(t - \Delta t))^2}}{\Delta t} \quad (2)$$

where $p(t)$ and $q(t)$ are neural activity vectors, $\Delta t = 20$ ms is the step size, and n is the number of dimensions in the component space. To create a distribution for multidimensional speed in different trial phases, we ran dPCA analysis 200 times, subsampling (20%, with replacement) neural firing rate trajectories for each run.

Neural decoding of firing rates

We pooled the firing rates of all neurons recorded across sessions and subjects to form a pseudopopulation of neurons. In this neural state space, we trained a decoder to perform population decoding for the prediction of correct/incorrect trials, set size of trials, and IN/OUT trials. The input was the matrix of neural firing rates of each trial of all neurons, binned in 1-ms windows, and smoothed in a 1500-ms window. The neurons were grouped according to anatomical location (hippocampus, entorhinal cortex, and amygdala). As the classifier, the decoder was trained and tested at 250-ms steps using a support vector machine as implemented in the NDT toolbox (25). Independently for each time point, classification was performed by assigning firing rates from randomly selected trials for each neuron to training and test splits in each of the 500 resampling runs. We first classified the trials based on workload (set size 4 versus set sizes 6 and 8) using 10 cross-validation splits. We selected units from sessions with at least five incorrect responses and then classified correct/incorrect trials using five cross-validation splits. To classify the IN/OUT trials after the probe, we used a narrow window size of 500 ms and 10 cross-validation splits. We tested significance by creating a null distribution of accuracies with 500 runs of the decoder with scrambled labels and comparing the true accuracy to the null distribution. This approach resulted in a minimum value of $P = 0.002$ for the statistical significance. We corrected the P value for multiple comparisons using Bonferroni's method. For the 12 tests performed (Fig. 4, D to F), the corrected value was $P = 0.002 \times 12 = 0.024$ and remained statistically significant.

EEG preprocessing

Data were visually inspected, and trials that contained eye movements or artifacts were removed using the toolbox EEGLAB (39).

Frequency spectra

To calculate spectra during maintenance (Fig. 5, B, C, E, F, and H), the data were segmented into 2-s epochs and grouped according to set size. We focused on the last 2 s of the maintenance period to avoid interference from visual evoked responses and eye blink artifacts. We used the multitaper method (± 2 Hz smoothing in the frequency domain with seven tapers) as implemented in FieldTrip (36) to reduce spectral leakage and control the frequency smoothing (36).

Phase-locking value

To probe the functional connectivity between the cortex and hippocampus, we calculated the PLV between the scalp EEG and iEEG channels (21, 36).

$$\text{PLV}_{ij}(f) = \frac{1}{N} \left| \sum_{n=1}^N \frac{X_i(f) \cdot (X_j(f))^*}{|X_i(f)| \cdot |X_j(f)|} \right| \quad (3)$$

where PLV_{ij} is the PLV between channels i and j , N is the number of trials, $X(f)$ is the Fourier transform of $x(t)$, and $(\cdot)^*$ represents the complex conjugate.

Using the spectra of the 2-s epochs, phase differences were calculated for each electrode pair (i, j) to quantify the interelectrode low-frequency phase coupling. The phase difference between the two signals indexes the coherence between each electrode pair and is expressed as the PLV. The PLV ranges between 0 and 1, with values approaching 1 if the two signals show a constant phase relationship over all trials (21).

To test the statistical significance of the PLV, we first permuted trials 200 times and computed the 95th percentile threshold within trials of set size 8. We used this one-tailed test for increase in the PLV at each frequency bin independently. Only the electrode pairs with PLV above threshold were selected for further processing. From the PLV spectra, we selected the electrode pair and frequency range of the highest PLV for each subject (table S1). Further analysis focused on these electrode pairs.

The workload-dependent increases in the PLV for the selected electrode pairs were subsequently assessed by subtracting the PLV for trials with set size 4 from the PLV for set size 8. The statistical significance was then estimated using a permutation test, in which a null distribution was created by randomly assigning trials (i.e., 2-s epochs) into two set size conditions, computing the PLV differences between conditions, and repeating this procedure 200 times. From the frequency spectra of the PLV differences for each subject, we considered only the frequency band of the PLV difference that was above the 95th percentile to be statistically significant (table S1). The edges of this individual frequency band (black bar in Fig. 5E) are presented in table S1, and further analysis focused on this individual frequency band.

Next, we determined the PLV difference between the two behavioral conditions encoding and maintenance by subtracting the PLV during encoding from the PLV during maintenance. Similar to the results above, the statistical significance was estimated using a permutation test, in which a null distribution was created by randomly assigning 2-s epochs into two behavioral conditions, computing the PLV differences between conditions, and repeating this procedure 200 times. Only the frequency points with PLV above the 95th percentile threshold were considered statistically significant. Thus, we could relate the PLV to the behavioral condition during task performance.

Time-frequency analysis

To calculate spectral power in the time-frequency domain (Fig. 5, A and G), each trial was transformed to a time-frequency map. We used multitapers with a window width of 10 cycles per frequency point, smoothed with $0.2 \times \text{frequency}$, and used three tapers (36). We computed power in the frequency range of 4 to 30 Hz with a time resolution of 0.1 s. The power during fixation ($[-6.0, -5.0]$ s) served as a baseline for the baseline correction ($\text{power} - \text{power}_{\text{baseline}} / \text{power}_{\text{baseline}}$).

for each time-frequency point. Using the same parameters, we computed the PLV in the time-frequency domain (Fig. 5D).

Spike-field coherence

We next quantified the coupling between the neuronal action potentials and the surrounding field. We calculated the spike-field coherence between a neuronal spike and the LFP recorded from the same micro-electrode. For this step, we used the pairwise phase consistency (PPC; $\hat{P}_2$) (21, 36, 40). Compared to PLV, this measure is more balanced toward differences in spike counts. To test the statistical significance of the PPC, we created a null distribution by permuting trials for spikes 200 times and computed the 95th percentile threshold. Using this threshold at each frequency bin, we computed the percentage of neurons with significant PPC (fig. S6).

SUPPLEMENTARY MATERIALS

Supplementary material for this article is available at <http://advances.sciencemag.org/cgi/content/full/5/3/eaav3687/DC1>

Fig. S1. Metrics for spike sorting for the identification of putative neuronal units.

Fig. S2. Examples of maintenance neurons.

Fig. S3. Details of the dimensionality reduction technique (dPCA) used for state space analysis.

Fig. S4. Frontal scalp EEG and PLV in subject 1.

Fig. S5. Scalp EEG power spectra for all subjects.

Fig. S6. Details of the spike-field coherence comparison between maintenance and encoding.

Table S1. Subject characteristics.

Table S2. Statistics reporting checklist.

Movie S1. Attractor dynamics in neuronal dPCA state space throughout the task.

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